

Stakeholder Engagement Plan

1. Selected Stakeholder Groups

Based on our stakeholder map, we select the two groups that span the full delivery of our retrieval system:

- **Smallholder Farmers:** These are the end users whose natural language questions drive the system and whose livelihoods depend on the quality of the retrieval advice.
- **Agricultural extension agents:** The field intermediaries who bridge the gap between technical knowledge and farmer practice and whose adoption decisions determine whether the system reaches the intended communities.

Together, these groups cover the demand side(farmers formulating real queries) and the delivery side(agents vetting and channeling retrieved information). Engaging both ensures that the retrieval engine is shaped by the people closest to the problem.

2. What Matters Most to Each Group:

Smallholder farmers care most about the accuracy and safety of the advice they receive: a wrong pesticide recommendation or an off-zone fertiliser regime can cause crop loss and financial hardship. Accessibility also ranks high-queries must be understood even when phrased in local languages or mixed codes, and answers must be usable by people with limited literacy or connectivity.

Extension agents prioritise efficiency and professional credibility. They need to find the right factsheet quickly during field visits, and they need the tool to reinforce, not undermine their expertise. Coverage matters too: if the knowledge base has blind spots on common local crops, the tool loses trust. Both groups share a concern for trustworthiness: the system should surface expert-validated content, not commercial advertising or outdated research.

3. Potential Harms and Benefits

	Potential Benefits	Potential Harms
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Smallholder Famers	Faster, more equitable access to expert-validated agronomic advice regardless of proximity to an extension office or internet connectivity. Empowerment through information: farmers make better-informed decisions on pest management, soil health, and climate adaptation. Reduction in the information gap between well-connected and marginalised rural communities.	Retrieval of advice designed for a different agroecological zone (wrong pesticide, wrong planting window) leading to crop damage or financial loss. Exclusion of underrepresented crops or regions from the corpus, systematically under-serving certain farming communities. Over-reliance on the tool without professional cross-checking, creating a false sense of certainty
Extension Agents	Efficiency gains during field visits: instant, ranked access to the right factsheet instead of manual search through PDF repositories. Extended reach: agents can cover more farmers with higher-quality, evidence-based advice. Professional development through exposure to a broader knowledge base than what they might individually command.	Perception that the tool threatens their professional role, leading to resistance or disengagement. Misrepresentation of complex agronomic advice when documents are retrieved out of context. Additional cognitive burden if the interface is poorly designed for low-resource field conditions (offline, low-bandwidth, small screens).

4. Engagement Methods from the Action Catalogue

With regards to the Engage2020 Action Catalogue (actioncatalogue.eu) and filtering for methods suited to participatory research with affected communities, small group sizes, and low-resource contexts, we selected the following methods:

- **For smallholder farmers:**

The method chosen is **preparatory design**(co-design workshops). Here, farmers contribute real queries and evaluate retrieval outputs face-to-face, ensuring the system reflects actual information needs.

- **For extension agents:**

We opt for the **User Committee approach**(standing advisory panel). Here, a small, recurring committee of agents provides structured, longitudinal feedback that directly shapes iteration cycles.

5. Why Each Method Fits

5a. Participatory Design for Smallholder Farmers

Participatory Design (PD) is a methodology rooted in the Scandinavian tradition of democratic co-creation, where the people most affected by a technology participate directly in shaping it. Rather than treating end-users as passive subjects of testing, PD positions them as experts of their own lived experience, an epistemological stance that is particularly important when building information systems for communities whose knowledge, language, and farming context are poorly represented in formal academic literature. The Engage2020 Action Catalogue lists PD as well-suited for projects aiming to surface ideas for new technology development and to involve affected communities in shaping research agendas, which is exactly the challenge our retrieval system faces. For our project, PD takes the form of co-design workshops conducted with small groups of smallholder farmers (8-12 participants per session) in their local communities. During these workshops, farmers are presented with a set of agricultural queries; some drawn from the competition training set, others formulated by the farmers themselves in real time and shown the top-5 documents that our retrieval system returns. Farmers then evaluate, in their own terms, whether the retrieved factsheets actually answer the question they had in mind. This is fundamentally different from asking an external expert to judge relevance: it captures the farmer's understanding of what constitutes a useful answer, which may diverge from academic judgment. For instance, a factsheet on maize nutrient deficiency might score highly on expert relevance labels, but a farmer whose maize variety is specific to her altitude and soil type may consider it irrelevant a discrepancy that only emerges through direct participation. This method fits our stakeholder group for several structural reasons.

First, smallholder farmers in Sub-Saharan Africa often operate in oral, communal decision-making cultures where knowledge is shared through dialogue rather than written forms. PD workshops, especially when facilitated in local languages, align naturally with this communication mode, lowering the barrier to meaningful participation.

Equally, PD does not require participants to be literate or digitally fluent. Farmers interact with printed or projected retrieval results, and facilitators read or summarise content aloud when needed. Visual aids such as crop photographs or colour-coded relevance cards can replace text-heavy evaluation forms.

Furthermore, PD generates qualitative data that no automated evaluation metric can capture: why a document feels trustworthy or misleading, what contextual information is missing, and what follow-up questions the farmer would ask. These insights feed directly into our retrieval pipeline. For example, by identifying hard negatives that should be penalised during re-ranking, or by surfacing crop-topic gaps that signal corpus-level bias.

A practical consideration is cost. PD workshops require physical facilitation, travel to farming communities, and potentially compensation for participants' time.

However, the method scales well when embedded within existing extension service visits or farmer cooperative meetings, minimising logistical overhead. The workshops also yield a secondary benefit: they build trust and ownership, making the farmer community more likely to adopt and champion the tool when it reaches deployment. The key risk is tokenism; running workshops that collect input but never act on it. We mitigate this by committing to a feedback loop: after each workshop round, we document which retrieval failures were flagged, implement re-ranking adjustments, and share the results in a follow-up session. This iterative, two-way engagement is what distinguishes genuine PD from superficial user testing.

5b. User Committee for Agricultural Extension Agents

A User Committee, as catalogued in the Engage2020 Action Catalogue, is a standing advisory group composed of representative stakeholders who meet regularly over the life of a project. Unlike one-off consultations, a User Committee provides sustained, structured engagement. Members develop familiarity with the system's evolution and can offer increasingly informed and nuanced feedback over successive iterations. The Action Catalogue identifies this method as particularly effective when the project needs ongoing input from a stakeholder group with moderate to high influence, which matches

the profile of extension agents in our stakeholder mapping ("Manage Closely" quadrant: high interest, high influence).

For our project, we propose a User Committee of 5-8 agricultural extension agents recruited from different agroecological zones covered by the corpus (e.g., humid forest, guinea savanna, highland). The committee convenes bi-weekly, alternating between in-person meetings and lightweight remote check-ins via phone or messaging platforms such as WhatsApp, which are already part of agents' professional routines. Each session follows a structured protocol: agents review a batch of recent retrieval outputs, annotate cases where the ranking failed or succeeded, and flag emerging query patterns from their field interactions. Over time, the committee builds a shared annotation corpus that serves as a qualitative complement to the quantitative $nDCG@5$ metric.

The fit between User Committees and extension agents is grounded in several characteristics of this stakeholder group. To begin, extension agents are trained professionals with domain expertise. They possess the agronomic vocabulary and critical judgment needed to evaluate whether a retrieved factsheet is not just topically related but practically sound for a given farmer context. This makes them ideal for detecting the subtle hard-negative failures that our Data Card identified as a priority, documents that look similar on the surface but address a different crop, zone, or growing season.

Moreover, agents occupy a gatekeeping position in the information delivery chain. A User Committee gives them a formal role in shaping the tool, which directly addresses the engagement risk identified in our stakeholder mapping: the perception that the system threatens their professional autonomy. By positioning agents as co-evaluators rather than passive recipients, the committee transforms potential resistance into active ownership.

Additionally, the longitudinal nature of the method is critical. Our retrieval pipeline is not a static product. It evolves as we experiment with different approaches, and each configuration change may introduce new failure modes. A standing committee can track these shifts over time, comparing system behaviour across iterations in a way that one-off evaluations cannot. This mirrors real-world software development practice, where beta-tester panels provide ongoing regression feedback. Fourth, extension agents are geographically distributed across the regions the corpus covers. A committee drawn from multiple zones provides a natural mechanism for detecting regional bias in retrieval performance; if an agent from the highland zone consistently flags poor results for altitude-sensitive queries, that signal is far more actionable than an aggregate $nDCG$ score averaged across all queries.

The practical feasibility of this method is high. Extension agents are already embedded in institutional structures (government extension services, NGO programmes) that facilitate recruitment and continuity. The bi-weekly cadence is light enough to respect their field schedules while frequent enough to generate useful data. The main challenge is attrition: committee members may be reassigned or lose motivation. We address this by keeping the committee small, recognising contributions publicly (e.g., acknowledgment in project documentation), and ensuring that every round of feedback leads to a visible system improvement, demonstrating that their input has real consequences.

6. Influence on Dataset and Model Design

Both engagement methods feed directly into our retrieval pipeline. Participatory Design workshops with farmers generate new, real-world queries that expose corpus coverage gaps: crops, pests, or agroecological conditions that the existing 695 factsheets do not adequately address, prompting targeted corpus expansion or the introduction of query-rewriting strategies to handle under-represented topics. The User Committee's ongoing annotations create a supplementary set of expert relevance judgments that can be used to fine-tune re-ranking models beyond the competition-provided qrels, particularly for hard-negative cases. Together, these feedback streams ensure that data sourcing, relevance labeling, and ranking model design are iteratively grounded in the needs and knowledge of the communities the system is meant to serve.